

ArenaMind

PRD (Product Requirements Document)

Part 1

ArenaMind AI

Product Requirements Document (PRD)

Version: 1.0

Project Name: ArenaMind AI

Tagline:

| *The Generative AI Operating System for FIFA World Cup Stadiums.*

Prepared By

Product Management Team

Target Event

FIFA World Cup 2026 Hackathon

Document Status

Draft v1.0

Table of Contents

1. Executive Summary
2. Problem Statement
3. Vision
4. Mission
5. Product Overview

6. Why ArenaMind AI
 7. Market Opportunity
 8. Product Goals
 9. Success Metrics
 10. Stakeholders
 11. User Personas
 12. Functional Modules
 13. AI Capabilities
 14. Product Scope
 15. Out of Scope
 16. User Journeys
 17. Functional Requirements
 18. Non-Functional Requirements
 19. MVP Definition
 20. Future Roadmap
 21. Risks
 22. Appendix
-

Executive Summary

Overview

The FIFA World Cup 2026 will be the largest sporting event ever organized.

Unlike previous tournaments, the 2026 edition introduces:

- 48 national teams
- 16 host cities
- 3 host countries

- Millions of international visitors
- Hundreds of thousands of spectators per match
- Thousands of volunteers
- Multiple transportation agencies
- Complex multilingual communication
- Massive real-time operational challenges

Although stadiums are increasingly digitized, tournament operations still rely on disconnected software platforms.

Current operational systems are fragmented across:

- Security operations
- Crowd monitoring
- Transportation
- Volunteer management
- Medical teams
- Sustainability monitoring
- Fan services
- Emergency response
- Public announcements

Each department typically operates in isolation, making real-time coordination slow and inefficient.

This creates operational blind spots during large-scale events, where every second matters.

ArenaMind AI addresses this challenge by introducing a unified Generative AI Operating System capable of orchestrating stadium operations in real time.

Rather than functioning as another chatbot, ArenaMind AI becomes an intelligence layer that sits on top of every operational system within the stadium ecosystem.

Using Generative AI, digital twin technology, predictive analytics, multimodal AI, and real-time event intelligence, ArenaMind AI transforms stadium management from reactive decision-making to proactive operational intelligence.

The platform continuously observes operational signals from cameras, IoT devices, transportation APIs, weather feeds, crowd sensors, ticketing systems, and volunteer networks to generate recommendations before issues escalate.

ArenaMind AI empowers:

- Stadium operators
- Security teams
- Event organizers
- Volunteers
- Transportation authorities
- Medical staff
- Sustainability teams
- Fans

through one unified AI interface.

Product Vision

To become the world's first Generative AI Operating System capable of orchestrating large-scale sporting events through intelligent automation, predictive decision support, and real-time operational intelligence.

ArenaMind AI envisions a future where every stadium operates like a living, intelligent ecosystem—able to understand, predict, and optimize itself.

Mission Statement

Build an AI-powered operating system that enables safer, smarter, more sustainable, and more enjoyable sporting events through real-time intelligence, multimodal AI, and predictive decision-making.

Problem Statement

Current Situation

Modern stadiums are equipped with numerous digital technologies.

Examples include:

- CCTV cameras
- Access control systems
- Smart ticketing
- Parking systems
- Transportation APIs
- Digital signage
- Security sensors
- IoT infrastructure
- Environmental monitoring
- Energy systems

Despite this technological maturity, these systems rarely communicate intelligently with one another.

Operators often switch between multiple dashboards during live matches.

A security operator may need to monitor:

- CCTV software
- Incident dashboard
- Radio communication
- Crowd monitoring
- Access control

simultaneously.

Likewise, transportation authorities monitor independent systems.

Medical teams operate independently.

Volunteers receive updates through messaging applications.

Fans rely on different mobile applications for navigation, food, transportation, and ticketing.

This fragmented ecosystem creates significant operational inefficiencies.

Key Challenges

1. Crowd Congestion

Large crowds frequently create bottlenecks near:

- Entry gates
- Food courts
- Washrooms
- Merchandise stores
- Metro stations
- Parking lots

Current systems only detect congestion after it has already formed.

There is limited ability to predict crowd movement before congestion becomes dangerous.

2. Fragmented Operational Intelligence

Operational teams lack a unified understanding of stadium conditions.

Information remains siloed.

Decision-making becomes reactive rather than proactive.

3. Poor Fan Navigation

Visitors frequently struggle with:

- Finding gates

- Locating food vendors
- Discovering accessible facilities
- Parking
- Public transportation
- Real-time route optimization

This reduces overall spectator satisfaction.

4. Multilingual Communication

The FIFA World Cup welcomes fans from hundreds of countries.

Emergency announcements, navigation instructions, and operational messages often require rapid multilingual translation.

Current workflows depend heavily on manual processes.

5. Emergency Coordination

During emergencies, multiple departments must coordinate simultaneously.

Examples include:

- Security
- Medical teams
- Volunteers
- Police
- Transportation
- Fire services

Existing systems provide limited AI-assisted coordination.

6. Volunteer Coordination

Thousands of volunteers operate throughout tournament venues.

Task assignment is often static.

Dynamic reassignment based on real-time operational changes remains limited.

7. Sustainability Optimization

Large sporting events consume enormous quantities of:

- Electricity
- Water
- Food
- Fuel
- Transportation resources

Optimization opportunities are frequently missed because systems cannot reason across multiple environmental datasets simultaneously.

Why Existing Solutions Fall Short

Current platforms generally solve individual operational problems.

Examples include:

Platform	Limitation
Crowd Monitoring	Only visualizes existing congestion
Transportation Systems	Operate independently
CCTV Platforms	Lack contextual AI reasoning
Help Desks	Reactive support only
Chatbots	Limited operational awareness
Volunteer Apps	Static workflows
Energy Monitoring	No predictive optimization

There is no unified intelligence layer capable of understanding the complete operational state of an international sporting event.

Proposed Solution

ArenaMind AI introduces a centralized AI Operating System that continuously observes, understands, predicts, and optimizes stadium operations.

Unlike conventional dashboards, ArenaMind AI combines:

- Generative AI
- Digital twin modeling
- Predictive analytics
- Real-time event processing
- Vision AI
- Natural language interfaces
- Operational knowledge graphs
- RAG-powered decision intelligence

to deliver actionable recommendations rather than raw data.

The platform enables users to interact with stadium operations naturally.

Example:

┆ "Show me all operational risks expected within the next 30 minutes."

ArenaMind AI automatically analyzes:

- Crowd density
- Transportation delays
- Volunteer availability
- Weather conditions
- CCTV metadata
- Incident reports
- Stadium capacity
- Energy consumption
- Medical availability

before generating prioritized recommendations complete with confidence scores and suggested mitigation actions.

Product Value Proposition

ArenaMind AI transforms stadium operations from reactive monitoring into proactive operational intelligence.

Instead of asking:

"What is happening?"

operators begin asking:

"What is likely to happen next?"

This shift enables:

- Faster decisions
- Improved safety
- Better fan experiences
- Reduced operational costs
- Lower environmental impact
- Improved accessibility
- Enhanced multilingual communication
- Greater operational resilience

ArenaMind AI becomes the central intelligence layer connecting every operational stakeholder through one AI-powered ecosystem.

Product Positioning

ArenaMind AI is positioned as an enterprise-grade AI Operating System for large-scale sporting venues.

It is not:

- A chatbot

- A dashboard
- A monitoring tool
- A ticketing platform

Instead, it is an orchestration layer that augments existing infrastructure using Generative AI and predictive intelligence.

ArenaMind